

Towards an informational account of interpersonal coordination

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Abstract

Human sociality is grounded in the dynamic coordination of individuals as they interact with one another. Indeed, various levels of interpersonal coordination – neural, behavioural, physiological, affective, linguistic – are hallmarks of successful social communication and cooperation. However, describing these complex, interdependent dynamics has been limited by current methodological approaches, owing to a restrictive repertoire of tools and the absence of a unified, standardized methodological framework. Here, we identify information theory – the mathematical theory of communication – as a particularly well-suited conceptual framework to address this shortfall, given its appropriate sensitivity to complex dynamics, including potential nonlinearity and higher-order interactions, and its data-driven, model-agnostic foundations. With deep roots in computational, cognitive and systems neuroscience, the formal introduction of information-theoretic quantities and methods into the study of interpersonal coordination is perhaps overdue. In this Perspective, we advance the case for a unified information-theoretic framework for the field while paving the way for a new generation of empirically testable, theoretically grounded research questions.

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Introduction

Human life is necessarily social. The cultivation of personality traits, dispositions, opinions, ethical commitments and repertoires of emotional, linguistic and gesticulatory responses are fundamentally shaped by the surrounding social environment. What are humans but an emergent mosaic of their social lives, facilitated and constrained by their biological bodies?

Despite the innate sociality of daily life, until recently, most psychological and neuroscientific investigations into human sociality have focused on the individual¹. However, in the past two decades, multimodal experiments and hyperscanning (the simultaneous neuroimaging of multiple, interacting persons) have offered promising avenues for exploring how people seem to coordinate themselves behaviourally, physiologically and neurally, for example, during social interactions^{2,3} (Fig. 1). We term this ‘multimodal interpersonal coordination’. Importantly, interpersonal coordination is not merely an epiphenomenon (a causally inert by-product) of interaction but is theorized to support fundamental social functions, including effective communication, joint action and cooperative problem-solving^{4–8}. Indeed, the alignment between interactants, whether brain-to-brain or cross-modal (such as linguistic-to-neural) relationships, has been frequently identified as a marker of successful social communication^{1,9}. ‘Being on the same wavelength’, then, has become more than a fanciful metaphor for interpersonal rapport and flow, but a neurobiological reality. Given the role of interpersonal coordination in everyday sociality and even human evolutionary history⁸, there is a clear and pressing need to grasp its mechanistic, dynamical and cross-modal foundations. In this Perspective, we argue that capturing the true complexity of interpersonal coordination requires moving beyond simplistic assumptions and towards

analytical approaches capable of engaging the rich, dynamic and multi-level nature of social interaction.

Social and neural systems are dynamical systems; that is, systems that are continuously evolving and whose components are often characterized by complex, interdependent relationships. Importantly, interpersonal systems can be investigated on many scales of spatiotemporal organization (across scales of space and time). Take the example of social movements. One could investigate the long-term diffusion and persistence of political ideologies, the coordinated behaviours of protesters who chant and march synchronously over several hours, or even the millisecond-level electrophysiological activity of co-partisans and political rivals as they debate one another.

Understanding the mechanisms behind interpersonal coordination has been a central challenge in social psychology and neuroscience. Here, we take coordination to be a type of integration, describing systems in which some sender A transmits ‘information’ to some receiver B (directly or indirectly) who, in turn, integrates that information such that they experience a tractable change (Fig. 2a). Crucially, this process need not be linear. In fact, complex coordination is often nonlinear, which means that responses may be disproportionate (non-proportional) or context-dependent in ways that resist decomposition into individual contributions (non-separable). For example, reactions to particular words can be non-proportional compared with their initial utterance, and words themselves ought to be understood as inseparable from their wider context. Similarly, the information transfer between teacher and student may be variable depending on student capability and attentiveness, and the social dynamics of their classroom may be better described as their emergent, totalizing effect rather than the cumulative behaviour of individual students. Thus, the study of sociality should not be limited to a single

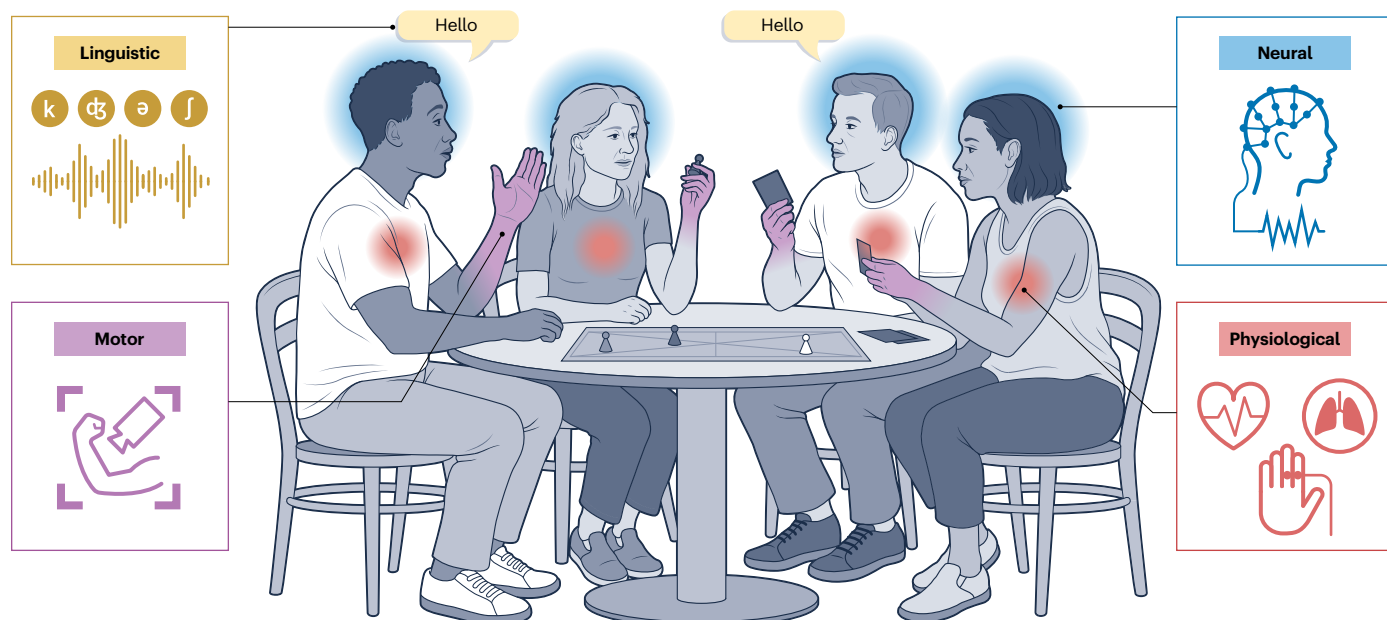
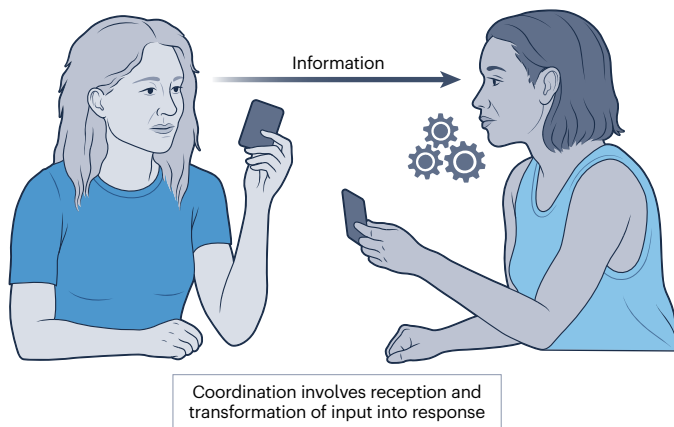


Fig. 1 | Multimodal interpersonal coordination. Interpersonal coordination of a social group (who may be conversing or engaging in a competitive or collaborative task) across different levels of organization. Non-exhaustively, these levels include the following: motor, which refers to hand, arm and eye movements (kinematic velocity and position can be recorded by motion-sensor cameras and eye-tracking systems); linguistic, wherein intonation,

prosody, acoustic and semantic features may be captured; neural, wherein the neural dynamics of participants can be recorded (most often via electroencephalography or functional near-infrared spectroscopy⁶); and physiological, which spans heart and respiration rate and galvanic skin response, for example.

a Defining interpersonal coordination

Interpersonal coordination = propagation + integration



b Signal transformations

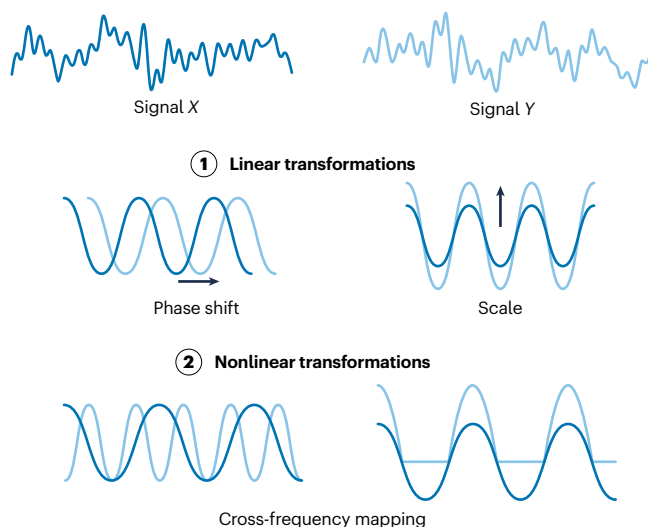
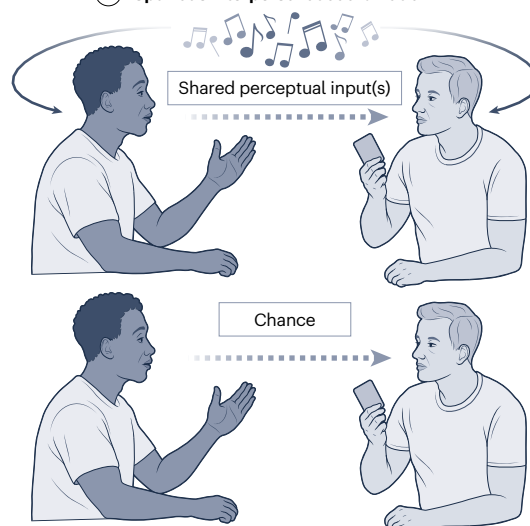


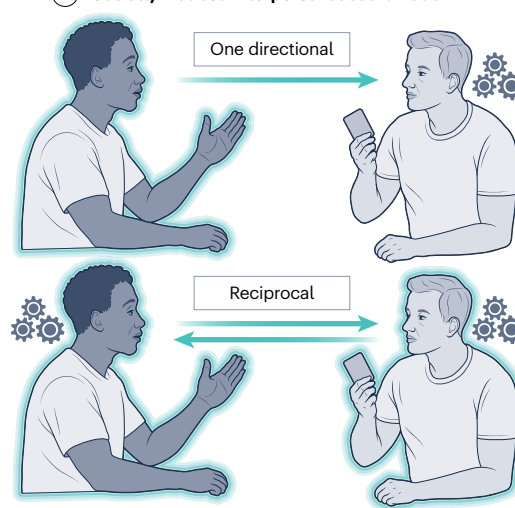
Fig. 2 | Defining interpersonal coordination. **a**, A definition of interpersonal coordination is proposed here based on findings from intra-brain, inter-areal communication. Interpersonal coordination requires more than signal propagation: whereas some forms may involve the receiver mirroring the input through proportional mapping, more complex forms involve the receiver modifying the input through internal dynamics, reflecting integration. Although propagation and integration may both involve linear or nonlinear transformations, distinguishing between them requires approaches that go beyond signal similarity. **b**, (1) Linear transformations detectable by traditional measures include scaling and phase shift, wherein output signals (Y) remain proportional and frequency-aligned with the input signals (X). (2) By contrast,

c Coordination types

① Spurious interpersonal coordination



② Socially induced interpersonal coordination



nonlinear transformations (such as cross-frequency mapping) reflect more complex, non-proportional dependencies between input and output signals that linear methods cannot capture. **c**, Distinguishing genuine from spurious interpersonal coordination, following ref. 79. (1) Apparent interpersonal coordination on the neural level may arise spuriously owing to shared environmental input (both partners hearing music) or coincidence (without any interaction, by chance). (2) By contrast, meaningful interpersonal coordination can emerge through directed (unidirectional) or reciprocated (bidirectional) socially induced information exchange. Although linear methods may detect shared signal structure, only analyses sensitive to transformation and integration can reliably distinguish between passive correlation and functional interaction.

resolution across space and time nor be described by fixed, additive models but instead should embrace approaches capable of capturing this nonlinear, distributed complexity of interactions across levels.

However, the field has yet to settle on a consistent approach to the multi-level analysis of interpersonal interactions, with a disparate and diverse set of methodological and analytic tools being deployed

ad hoc (see refs. 3,10 for reviews of current methods and measures). Myriad measures are used to investigate different aspects of social interactions, making both cross-modal analyses (for example, interpersonal coordination across linguistic and neural levels of organization; Fig. 1) and cross-study comparisons difficult to interpret (raising replicability and generalizability concerns)^{3,10,11}. Moreover, although

the importance of nonlinear and higher-order interactions and conducting cross-modal analyses is acknowledged, such approaches are largely absent in the current literature. Despite promising efforts in methods development, contentious methodological decisions linger (see 'Perils' section), and the exact mechanisms that drive multimodal interpersonal coordination remain contested³.

In this Perspective, we begin by highlighting both the current promises and perils of the field of interpersonal coordination and propose information theory as a particularly well-suited methodological framework to address existing shortcomings. We then provide an accessible introduction to various information-theoretic measures and to the opportunities that this methodological paradigm could confer for social psychologists and neuroscientists in generating novel, programmatic research on empirical questions that are central to the study of interpersonal coordination. Last, we present testable predictions including an illustrative experimental set-up and explore how our framework could help to re-address ongoing debates pertaining to the nature of interpersonal coordination.

Promises

In the pursuit of understanding interpersonal coordination, social psychology and neuroscience initially relegated themselves, somewhat ironically, to studying the individual – using simple, well-controlled paradigms such as the passive observation of noninteractive, artificial social stimuli (such as schematized faces) or using confederates (aides posing as participants) to uncover the neural mechanisms of social cognition^{12,13}. Although meeting stringent criteria for experimenter control, these approaches grounded in the tradition of experimental psychology necessarily fell short of capturing the dynamic, reciprocal and richly textured nature of human social interactions^{4,14–17} – the so-called dark matter of social neuroscience^{9,18} – as they typically involved one-sided, noninteractive tasks lacking mutual contingency. In response to these challenges, an 'interactive turn' took root, aimed at developing ecologically valid and naturalistic studies of socially engaged participants, rather than passive or simulated interactions.

Among the methods that emerged, the hyperscanning paradigm presented itself as an attractive solution¹⁹, involving the deployment of multiple, often identical, neuroimaging and/or electrophysiological technologies to simultaneously record the neural activity of two or more brains²⁰. For the past two decades, neuroscientists have simultaneously tracked the real-time neural dynamics of interacting participants via hyperscanning, as well as other levels of interpersonal exchanges on linguistic, behavioural, affective and physiological dimensions^{3,6} (Fig. 1). Notably, experiments using this 'second-person'¹⁶ or 'relational'²¹ approach yield a common finding: the phenomenon of multimodal interpersonal coordination. Here, we advocate for the term 'coordination' over 'synchrony' as the latter, often limited to time-locked events, represents only a subset of the former.

In studies of naturalistic social interactions, hyperscanning research consistently identifies brain-to-brain synchrony, 'alignment', or 'coherence' as key outcomes driven by interpersonal factors (see refs. 3,6,9,11,21,22 for detailed reviews). The driving interpersonal factors include familiarity^{23–25}, mutual attraction²⁶, aligned affective states^{27,28}, cooperative or competitive contexts²⁹ and social cues^{6,30}. In addition, neural coordination is correlated with verbal^{31–34} and non-verbal^{30,35,36} coordination. For example, interpersonal coordination at the neural level is associated with mutual gaze^{23,30,37,38}, changes in pupil dilation^{11,39}, postural sway⁴⁰, and synchronized breathing and heart rate patterns¹¹.

It is important to recognize that although interpersonal coordination occurs across multiple levels of organization (Fig. 1), some are clearly observable by one's interaction partner (speech, movement, breathing) but others are not (heart rate and neural activity, but see ref. 41 for evidence that individuals can partially infer heart-beat information by visually observing the face of another person). Overt signals drive interpersonal adjustments, with some cues being rendered consciously (gaze, speech) and others subconsciously (pupil size, breathing rate), as interactants rely on these perceptual cues to regulate and align themselves (voluntarily or involuntarily) in real time³⁷. Unobservable neural and physiological dynamics may, in turn, influence and be influenced by these behavioural exchanges, forming a multi-level feedback loop that underpins successful interpersonal coordination^{42,43}. Indeed, questions about the directionality and dynamics of these interdependencies remain⁴⁴. For example, these multimodal features may act as antecedents and/or consequences of interpersonal coordination at the neural level, or indeed they may dynamically and reciprocally influence each other. Collectively, these studies also highlight the probabilistic, multi-faceted nature of multimodal interpersonal coordination, whereby the likelihood of interactive success is predicated in a nonlinear fashion on a combination of input cues and contextual factors.

More recently, Gordon and colleagues⁴⁵ introduced an extensive and compelling theory of flexible multimodal synchrony, setting the unit of analysis at the social unit (such as an interacting dyad) and identifying synchrony as an emergent coordination of co-occurring changes between interactants across different levels of organization. Although acknowledging the dynamical and multi-stable nature of such interactions, building from work in dynamical systems^{46–50}, they similarly highlight a lack of standardized measures and the varied meanings and associations of synchrony across different scales of organization⁴⁵. Recent empirical work by Wu and colleagues⁵¹, although not involving neural data, demonstrates how fine-grained behavioural tracking in open-ended virtual environments can uncover real-time adaptive individual learning, social learning and imitation. Such work aligns with proposals that inter-brain plasticity (the change in inter-brain coordination over time) underpins social learning by facilitating the development of shared inferential models through repeated interactions^{7,52,53}. Advances in machine learning and natural language processing approaches have also been used to examine the neural and linguistic 'convergence' and 'divergence' among strangers and friends, respectively, in conversation⁵⁴, with participants' dynamic neural representations of cognitive and affective states decoded and tracked across tasks, particularly in relation to relationship type, conversation content and interaction goals. The challenge facing social neuroscientists is not one of data scarcity but making sense of the data.

Nonetheless, current statistical approaches may fail to capture the full richness of these complex, cross-modal dynamics of interpersonal systems. For multimodal data, Hamilton⁵⁵ recommends using a general linear model, wherein the neural dynamics of one person are modelled as a linear combination of the neural and behavioural data of that person's partner, as well as that person's own behaviour (for example, see ref. 38). By further incorporating random effects to account for dependencies in multimodal data, linear mixed models have been and continue to be deployed, too^{56,57}. Nonetheless, for both modelling approaches, given the potential nonlinearity and complex interdependencies characteristic of neural and social systems, assumptions that these dynamics can be represented by simple linear combinations of independent processes might be overly reductive.

In a different vein, Reindl and colleagues²⁵ used a Bayesian hierarchical model⁵⁸ for their multimodal data analysis, finding that although mother–child dyads exhibited higher neural synchrony than adult stranger–child relations, such synchrony remained statistically differentiable from autonomic nervous system or behavioural synchrony. Notably, after comparing models with Gaussian and nonlinear (lognormal) response distributions, they found that the nonlinear response distribution provided a better fit for their synchrony metrics²⁵. Recognizing the interdependency of multimodal interactions, Horowitz-Kraus and Gashri⁵⁹ introduced a synchronization matrix that systematically maps the pairwise interaction relationships between synchrony modalities (including neural, motor, gaze, acoustic and heart rate), providing a structured comparison of their interactions. More recently, Gashri and colleagues⁶⁰ used diffusion maps to integrate electroencephalography (EEG) hyperscanning, joint attention and motion analysis, emphasizing the continued importance of integrating and analysing different social interaction modalities. Nonetheless, for both latter examples, little attention was given to how ‘synchronization’ is measured between and across modalities, with all focusing on synchrony rather than coordination in general. The need for standardizing multimodal interpersonal coordination measurement and interpretation remains.

In all, although these approaches hold great promise for advancing our understanding of interpersonal coordination dynamics, profound challenges remain. Addressing the methodological limitations of the field will be crucial to fully realizing the potential of hyperscanning and multimodal analyses. We now turn to these limitations, examining how fragmented and assumption-driven methods risk distorting, rather than illuminating, the dynamics of interpersonal coordination.

Perils

In this section, we discuss two key limitations that we have identified in the current literature. We first document the perils of a fragmented set of analytic measures underpinned by assumptions about data structure, interaction dynamics and coordination mechanisms that may not hold in naturalistic settings. In addition, we identify contentious methodological practices in spectral analysis that may obscure, rather than reveal, the dynamics of interpersonal coordination.

Attending to interpersonal coordination, the hyperscanning literature hosts a large repertoire of measures, particularly those assessing phase and coherence relationships between time-series signals (see ref. 10 for review). However, many of these are limited by restrictive assumptions, such as Gaussianity (that data are distributed normally) and stationarity (that statistical properties do not change over time), with strict sensitivity to linear relationships. However, given that naturalistic paradigms often unfold in nonlinear, non-stationary and dynamic ways^{11,52}, these statistical analyses and their assumptions may pose problems^{61,62}. Indeed, nonlinearity is a key feature of neural systems^{63,64}, and using linear measures can often ignore nonlinear neural interactions (see refs. 62, 65–67 for examples). The critiques similarly apply not only to neural but also to any level of interpersonal coordination⁶⁸. Although linear methods have shown utility in specific cases^{69–71}, their misalignment with real-world naturalistic dynamics can introduce artefactual patterns or obscure meaningful dependencies^{65,72,73}. These challenges call for more flexible, assumption-free statistical approaches that remain sensitive to the inherent complexity of the data at hand.

Moreover, in interpersonal coordination between multiple people, simple correlation measures are insufficient to distinguish genuine

interactive coordination (whether directed or bidirectional) from spurious coordination that may arise from shared perceptual inputs or coincidental neuroanatomical similarities. Crucially, such measures cannot uncover the underlying causal mechanisms, making it unclear whether observed correlations reflect true inter-brain communication or merely common influences such as a shared acoustic environment^{2,55,74–80} (Fig. 2c).

Long-standing critiques have highlighted the above problems^{55,78,79}, although a unified alternative framework has yet to emerge. In addition, most linear methods do not capture the fluidity of social interaction, marked by dynamic transitions between periods of synchrony and asynchrony, characteristics of metastability observed in dynamical systems^{2,52,81}. Time-resolved measures, preserving temporal structure and tracking temporal variation at various levels of granularity ought to help to this end. Finally, higher-order approaches capable of addressing multivariate interdependencies have scarcely been applied, with one notable exception⁸². Although this may be enough to motivate a unified methodological framework, we present further critiques that have remained largely absent from the interpersonal coordination literature.

Social neuroscience, particularly in hyperscanning analyses, has demonstrated a continued reliance on spectral analyses grounded in the mainstream communication-through-coherence hypothesis, which posits that effective communication between brain areas is mediated by rhythmic, oscillatory synchronization⁸³. This approach has been implicitly (although sometimes explicitly⁸⁴) extended to systems of interacting brains⁷⁵; that is, physically distinct but socially engaged brains. Here, inter-brain coordination is assumed to be reflected in coupling between signals oscillating either at the same frequency or via cross-frequency interactions^{85,86}, wherein information transfer is thought to be most efficient⁸³. Although investigating the brain as an oscillatory system has been an extensive and fruitful research programme in neuroscience^{83,87–90}, this framework has become increasingly contested for investigating coordination even within the single brain. Thus, we argue that extension of the communication-through-coherence hypothesis to interpersonal systems is necessarily contentious on both theoretical and methodological grounds.

Recent experimental and theoretical work has challenged the communication-through-coherence hypothesis by showing that inter-areal interactions from a sender to a receiver, as typically measured in the brain using coherence or Granger causality, may simply reflect oscillatory power in the sender, rather than dynamic mutual entrainment or integration with the receiver^{65,91–93}. In this view, coherence appears to be the result rather than the cause of coordination, arguably reflecting structural alignment or passive propagation rather than functional connectivity^{91,92,94}. Linear, oscillatory interactions may therefore arise from the correlation of a sender with its own output projected to the receiver, leading linear measures such as coherence to spuriously infer coordination by conflating statistical dependence with true functional interaction^{65,91}. Put simply, just as speaking into an empty room does not constitute communication, signal propagation without integration cannot be taken as evidence of meaningful interpersonal coordination (Fig. 2a). Crucially, this reflects a fundamental limitation of linear metrics: they cannot distinguish genuine information exchange from shared signal structure or power fluctuations, especially in systems shaped by nonlinear or context-sensitive transformations^{65,91} (Fig. 2b).

By contrast, emerging perspectives on neurodynamics emphasize the potential nonlinearity of inter-areal coordination between

brain regions, suggesting that traditional linear models of functional connectivity may be insufficient to capture the complexity of neural interactions^{65,95,96}. According to nonlinear connectivity frameworks, interactions between brain areas may involve complex dependencies, including cross-frequency coupling, which are not well-captured by linear measures focused on same-frequency phase alignment^{65,93} (Fig. 2b). Empirically, nonlinear connectivity measures based on information theory outperform linear ones in capturing the long-range integration of sensory-relevant information across species, modalities and tasks^{97–104}. Despite this, linear oscillatory measures (particularly coherence within specific frequency bands) remain the most extensively used in hyperscanning analyses¹⁰. Compounded by the lack of direct inter-areal anatomical pathways between socially engaged brains, how oscillatory coherence, for example, should be interpreted between brains is increasingly unclear.

In addition, persistent methodological issues exist in how spectral features are derived, particularly for systems in which high temporal resolution is being recorded (such as by EEG or optically pumped magnetometer-magnetoencephalography¹⁰⁵) and, to a lesser extent, slower technologies such as functional near-infrared spectroscopy (fNIRS) and functional MRI (fMRI). Rhythmic neural oscillations are understood to manifest as distinct spectral peaks (or ‘periodic’ components) in the frequency domain. By simply plotting the power spectral density of the neural data and identifying putative spectral peaks, canonical frequency bands of interest can then be suggested to be implicated in the task at hand (Supplementary Fig. 1). These bands are often defined a priori – namely, delta (δ ; 1–4 Hz), theta (θ ; 4–8 Hz), alpha (α ; 8–13 Hz), beta (β ; 13–30 Hz) and gamma (γ ; 30–50 Hz) – and such exploratory investigations, including plotting power spectra and identifying spectral peaks, are scarcely reported. However, neural power spectra are characterized not only by the oscillatory peaks that sit atop but also by ‘background’ 1/f-like activity (or the non-oscillatory, ‘aperiodic’ component)^{65,106,107}. Because the aperiodic component contributes power across all frequencies with a characteristic slope and offset, it can artificially elevate baseline power within any frequency band. This elevation can mimic or exaggerate spectral peaks, thereby inflating estimates of oscillatory power and misleading interpretations of changes in rhythmic activity.

Thus, without accounting for this aperiodic component, standard transformations to the frequency domain (including Hilbert transform, Fourier transform and wavelet) may confound estimates of centre frequency and full frequency bandwidths, and even risk waveform distortions^{108–111}. Furthermore, practices of narrowband filtering (which typically assume that neural power within a predefined frequency range reflects oscillatory activity) ultimately fail to account for the underlying 1/f-like structure of the signal, may not necessarily extract putative oscillations and may also erroneously suggest the existence of spectral peaks when there are none^{109,112,113}. Although often neglected in hyperscanning analyses, the non-oscillatory component is an important feature of neural profiles in individuals that can vary with age, task demand and even state of consciousness (see refs. 113–118 for examples), and it can thus confound observed differences in periodic components between tasks or participants, leading to false inferences about interpersonal coordination on the neural level^{111,112}.

We explicitly suggest approaching interpersonal systems agnostically, forgoing assumptions on data structure and interaction relationships, while being equipped with a robust arsenal of measures and methodological decisions that are appropriately attuned to the

complexities of one’s data. For spectral analyses, in line with recommendations from Zamm and colleagues¹¹⁹, we suggest that analyses begin by separating oscillatory power from aperiodic activity via spectral parameterization, wherein estimating the 1/f exponent of the latter and obtaining the frequency, power and bandwidth parameters of participants become important preliminary analyses for disentangling their relative spectral contributions^{109,111,118,120} (Supplementary Fig. 1). However, care is needed to avoid statistical circularity (‘double dipping’), wherein the same data set is used both to define a feature (such as a frequency band) and to analyse it. Cross-validation, using separate subsets of data for feature definition and its subsequent analysis, and replication on independent data are recommended to ensure robustness (this caveat applies broadly to any data-driven feature selection, not only spectral analyses).

Finally, we note that the mutual prediction hypothesis may provide a viable account of how interpersonal coordination is achieved without invoking cross-brain communication mechanisms per se. Namely, it may help to explain inter-brain similarity as the result of agents independently predicting and adapting to the other’s behaviour rather than through direct interpersonal coordination on the neural level¹²¹. In the same vein, some have examined interpersonal dynamics across modalities through the computational lens of prediction error minimization⁵². Specifically, from a predictive processing framework, individuals align by minimizing prediction errors across hierarchical generative models, cascading from low-level predictions of movements to high-level predictions of cognitive representations^{52,122,123}.

In summary, the study of interpersonal coordination has increasingly acknowledged the complexity of interpersonal systems, although, many approaches still rely on simplifying assumptions about the nature of interpersonal coordination, data structure and spectral analyses. These assumptions, although sometimes necessary for practical and analytical reasons, may also limit the ability to fully capture the potentially nonlinear and dynamic nature of social interactions. Recognizing these trade-offs can help to advance more integrative approaches that balance theoretical complexity with methodological feasibility. Information-theoretic measures, to which we now turn, address the discussed limitations and we believe are well-suited to capturing the complexity of interpersonal coordination. With the firm grounding of information theory in computational, cognitive and systems neuroscience, the introduction of these quantities to the study of interpersonal coordination is perhaps overdue.

Proposal

On various scales of organization, one observes coordination whose unit of transfer can be loosely described as information. At first, defining information may not be entirely intuitive, let alone measuring its content (that is, what specific feature or property the information pertains to, such as size, shape or salience) or its distribution, transfer, storage, modification or other qualitative dynamics. Information theory – defined and refined by Claude Shannon and later lauded as the mathematical theory of communication^{124,125} – lends itself well to this cause, with information as its ‘common currency’^{126,127}. Grounded in probability theory, information theory offers a precise definition and quantification of information, particularly one that is content-invariant (not restricted by the nature of the content) and applicable to any data type including continuous time series, categorical data and discrete events, among others.

As we will demonstrate, information theory offers a powerful, data-driven and model-free approach to describing complex systems.

Namely, instead of making restrictive, statistical assumptions about multimodal data or forcing data to fit a pre-defined model, information theory allows the data set to ‘speak for itself’¹²⁸. This approach promises greater accuracy given its sensitivity and flexibility to data, which are particularly important for complex interpersonal systems. Indeed, some hyperscanning researchers have already noted the benefit of using information theory^{13,79} with a handful of empirical applications of information-theoretic measures^{82,129–138}. The formal introduction of information theory to the study of interpersonal coordination has yet to be explicated. We present this here.

Dependence and directional measures

Information is defined as a reduction in uncertainty (or ‘entropy’)^{124,139}. Simply, if knowing some ‘thing’ about variable X helps one to understand or predict variable Y better, one might say that X provides information about Y , as there is a reduction in uncertainty about Y upon knowing X . Put differently, the probability of Y behaving in a certain way, given one knows X (which is informative about Y), has increased. It follows that information has an inverse relationship with the probability of an event happening: the more probable something is, the less

uncertainty there is about it to be reduced (equation (1)). Roughly, if one can know the uncertainty or entropy about variables X and Y (denoted $H(X)$ and $H(Y)$), then the mutual information (MI) between them ($I(X; Y)$ in equation (2)) is the uncertainty in each variable that is resolved upon knowing the other variable (meaning that MI is symmetric; see Fig. 3a for visual heuristic):

$$H(X) = - \sum_x p_X(x) \log_2(p_X(x)) \quad (1)$$

$$I(X; Y) = H(X) - H(X|Y) \quad (2)$$

MI is an information-theoretic measure of statistical dependence between two random variables (such as X and Y) and can be understood as a generalization of Pearson’s correlation coefficient¹³⁹. Although ignorant to the direction of correlation (positive or negative), MI is sensitive to nonlinear interactions between variables. One can also remove any information about the statistical interaction that is also known via additional variable(s) (known as ‘conditioning’), analogous to partial correlation in the Pearson framework. In addition, MI can be

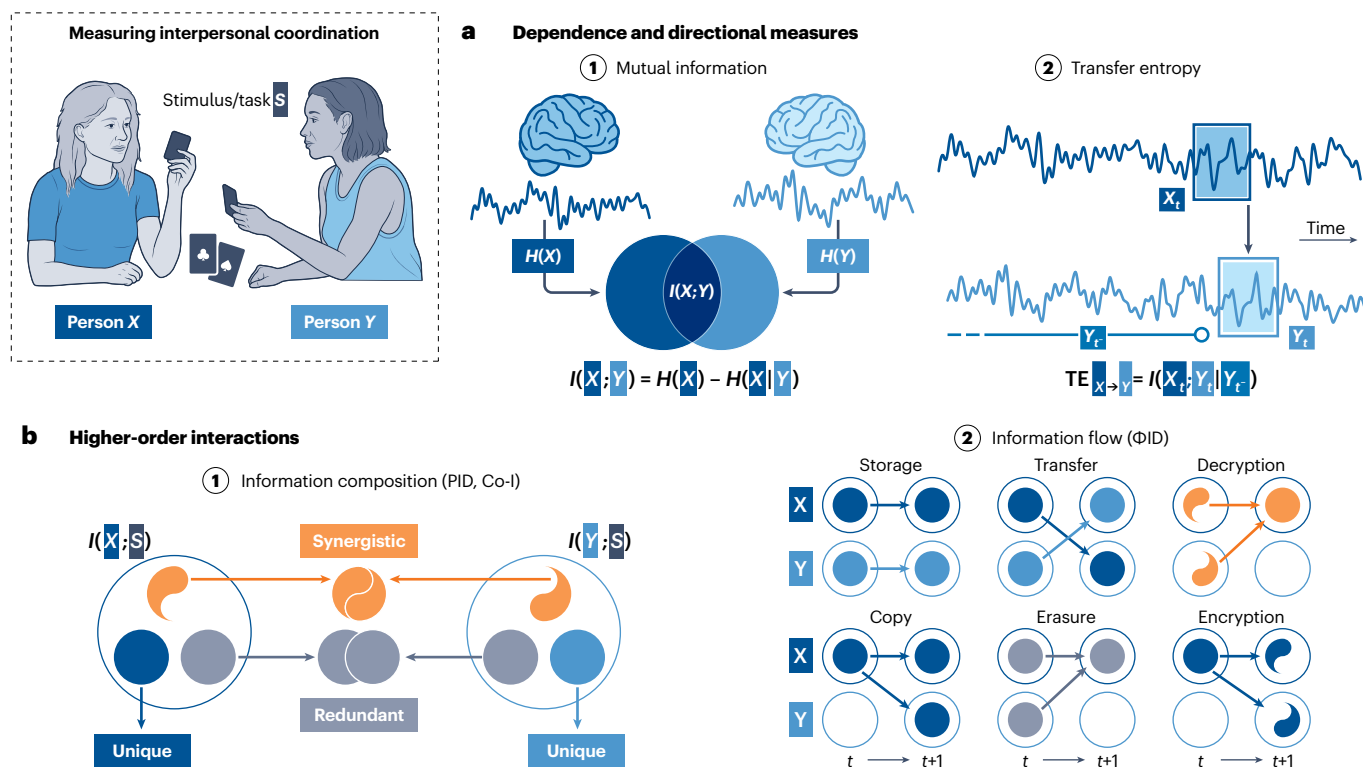


Fig. 3 | Proposed information-theoretic measures. Assuming the dyadic case, various information-theoretic measures can measure the interpersonal coordination between person X and person Y , engaged in a task or in perceiving stimuli S . **a**, (1) Mutual information is calculated using the Shannon entropies, $H(\cdot)$, of signals (in this example, neural, but could be any modality; see Fig. 1) and finding their intersection ($I(X; Y) = H(X) - H(X|Y)$). (2) Transfer entropy (TE) investigates how one time series may predict another based on the history of the latter ($TE_{X \rightarrow Y} = I(X_t, Y_t | Y_{t-1})$). **b**, Higher-order information-theoretic measures extend beyond pairwise dependencies to capture how information is jointly represented across multiple variables. (1) Partial information decomposition (PID) and co-information (Co-I) decompose the total information that sources X and Y

share about an external variable or stimulus S into ‘redundant information’ (shared by both sources, for example, when both encode the same visual cue), ‘unique information’ (available in only one source, for example, when X alone encodes emotional tone), and ‘synergistic information’ (emerging only when sources are combined, for example, when gaze and speech together reveal an intention that neither alone conveys). (2) Integrated information decomposition (ΦID) generalizes these higher-order approaches to dynamic systems with multiple sources and targets, tracking how unique, redundant and synergistic information evolve over time to quantify modes of information dynamics such as storage, transfer, copy, erasure, decryption and encryption.

resolved both temporally and spectrally to obtain time-domain and frequency-domain features of data sets¹⁴⁰ (see refs. 130,137 for nascent applications in hyperscanning research). Moreover, MI can measure the strength of effect sizes (given in bits) to facilitate cross-modal and cross-study comparisons^{128,141}.

Although MI is symmetric (like Pearson correlation), in a scenario that involves time, one may wish to reveal the directed information transfer between one time-series variable and another. This can be quantified by the transfer entropy (TE). TE can be understood as a conditional extension of MI in which the MI between time-series variables X_t and Y_t is conditioned on the history of the target variable Y_{t-} (refs. 139,142) (equation (3) and Fig. 3a). In other words, TE answers the question: “what could X predict about Y that the history of Y could not already predict?”

$$TE_{X \rightarrow Y} = I(X_t; Y_t | Y_{t-}) \quad (3)$$

TE may also be understood as a nonlinear generalization of the popular Granger causality measure, and indeed they are mathematically equivalent when assuming Gaussianity, thus bridging autoregressive and information-theoretic approaches in statistical causal inference¹⁴³. For comparison, X is said to Granger-cause Y if knowing X improves one's ability to predict the future of Y . However, TE proper (as defined in equation (3)) is model-free and, thus, does not assume the relationship type between variables, rendering it more versatile to apply to systems in which the underlying dynamics remain unknown. TE has already found some applications within the hyperscanning literature. For example, Liu and Pelowski¹³⁵ first explicitly recognized the substantial value of TE for quantifying information propagation between brains, including their causal direction, whereas on behavioural scales, both Trendafilov and colleagues¹³⁶ and Takamizawa and Kawasaki¹⁴⁴ were able to identify leader (information sender) and follower (information receiver) roles in joint motor coordination tasks.

In summary, although MI provides a model-free measure of statistical dependence, TE provides a powerful, directional development of MI and addresses the shortfalls of static descriptions of the information content of systems^{142,145–147}.

Higher-order interactions

To determine exactly how information (the reduction in uncertainty) about a target variable is encoded among multiple source variables, one can use partial information decomposition (PID)¹⁴⁸. This follows the intuition that there may be information held between multiple source variables that can synergistically contribute to the total joint information; in other words, information available in the whole may be more than the information available in the union of individual parts¹⁴⁹. Formally, given two (or more) source or predictor variables, X and Y , and one target variable, S , one can quantify how much information the sources (or predictors) provide about the target, both jointly, $I(X, Y; S)$, and individually, $I(X; S)$ and $I(Y; S)$.

From this, one can identify which shared information about a target variable may be redundant information (Rdn information), unique information (Unq information), or synergistic information (Syn information) (Fig. 3b). Rdn information is available from each source variable equally, such that it would still be present in the system even if one of the sources were missing, whereas Unq information is specific to a single source variable. By contrast, Syn information is learnt only from the conjunction of multiple variables and not any one of them individually (equation (4)):

$$I(X, Y; S) = \text{Syn}(X, Y; S) + \text{Unq}(X; S) + \text{Unq}(Y; S) + \text{Rdn}(X, Y; S) \quad (4)$$

In all, PID can be summarized as being concerned with decomposing information to uncover “who knows what about the target variable”¹⁵⁰.

Co-information (Co-I; or interaction information) is a multivariate information-theoretic measure that quantifies the shared information among three or more variables, capturing how the information shared collectively differs from what is shared pairwise^{151,152} (equation (5)):

$$I(X; Y; S) = I(X; S) + I(Y; S) - I(X, Y; S) \quad (5)$$

In other words, Co-I is a multivariate generalization of MI. However, unlike MI, the sign of the result (positive or negative) is indicative of whether the shared information is Rdn or Syn. Co-I can be extended to n -dimensions, too^{153–155}.

Finally, integrated information decomposition (ΦID) provides an extension of PID that can take into account multiple targets and multiple sources, tracking the dynamics of how Syn, Rdn and Unq information evolves over time^{156–158} (Fig. 3b). Simply, by decomposing the informational dynamics in these dynamic systems, distinct qualitative dynamical modes can be quantitatively derived, including information storage, duplication, erasure, transfer, and decryption or encryption. PID and its generalizations, including ΦID, have recently helped to identify higher-order relationships that may exist in the human brain^{159–161}, such as finding highly synergistic coalitions of brain regions that standard functional connectivity measures (Pearson correlation) would be blind to¹⁶².

Together, these measures offer a principled way to disentangle how information is distributed, shared or uniquely combined across interacting variables, revealing higher-order dependencies that conventional pairwise metrics cannot capture.

A universal statistical language

In all, information theory offers a comprehensive framework for characterizing the dynamics of information in dynamical systems. Whether information – defined as a reduction in uncertainty – is shared between variables or better predicted by another variable, MI and TE, respectively, represent two of the most fundamental measures of information dynamics and lend themselves well to the study of both dependency and directed transfer for any data type. Despite the former being ignorant to the direction of dependency, both measures are sensitive to nonlinear dynamics and do not require the expensive assumptions that are typically held by the current repertoire of measures.

For neural time-series data, MI and TE can be understood as extending the notion of ‘functional connectivity’ that is popular in cognitive and systems neuroscience (see ref. 163). Similarly, both MI and TE can be used to analyse the interaction between different modalities, such as motor and neural dynamics, that might inhabit different temporal resolutions (see case study in ref. 164 for an example). Higher-order measures such as PID, Co-I and ΦID, conversely, can uncover how various stimuli (whether from a shared environment or stimulus or indeed the social cues from the other interactant; Fig. 2c) are encoded between brains. Indeed, one might start identifying what type of information is being transferred or shared between brains or, in other words, what the computed information is about. For example, using Co-I, one can quantify the information about a stimulus variable that is encoded by two interacting signals (for example, across brains)¹⁰⁰. Decomposing

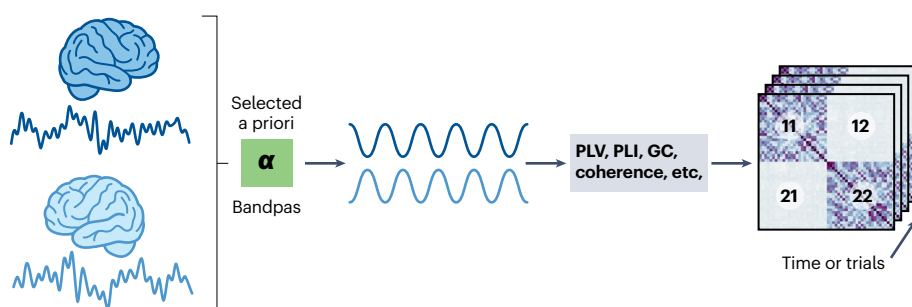
these dynamics into different qualitative modes and exploring how Rdn, Unq or Syn information can be held among multiple variables and about different stimuli are important steps forward for the study of interpersonal coordination (Fig. 3b).

A particular advantage of information theory as a mathematical framework for social psychologists and neuroscientists studying interpersonal coordination is its versatility and domain generality: the main unit of measurement is bits, an invariant quantity that applies to all data types, providing a unified language for diverse phenomena¹³⁹. In other words, information-theoretic measures offer a universal mathematical framework for statistical inference and analytic tractability, wherein direct evaluations of effect sizes (given in bits) can be made not only across diverse data types^{128,141} but also over both time and trials, wherein most standard measures are restricted either to one or the other¹³² (Fig. 4). Furthermore, these measures can be applied across neuroimaging technologies, experiments and even species¹⁶². This will provide a critical contribution towards efforts in replicability and generalizability.

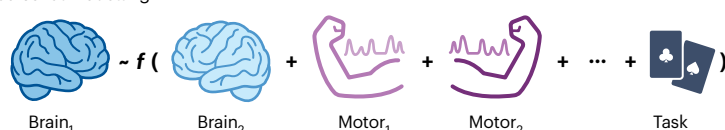
Extensions and developments in both temporal and frequency domains have facilitated exciting new information-theoretic approaches that are well-suited to complement and integrate seamlessly into current practices^{140,158,165,166}. As noted by Morimoto and Minagawa¹³⁰, in contrast to regression-based methods, which rely on predefined explanatory variables that are often event-based, subjectively selected and composed of multiple interdependent features, information-theoretic approaches offer a data-driven alternative. Information-theoretic methods identify statistical dependencies directly from the data, capture both linear and nonlinear relationships, and track evolving coordination dynamics without requiring discrete event segmentation. Crucially, unlike regression-based methods, they do not assume fixed unidirectional relationships and can instead capture bidirectional influences, making them particularly well-suited for investigating the complex, causal interactions during interpersonal coordination between the neural and behavioural levels wherein conclusions have hereto remained equivocal.

a Current approach

1 Neural coordination or 'synchrony'



2 Interpersonal modelling



b Proposed approach

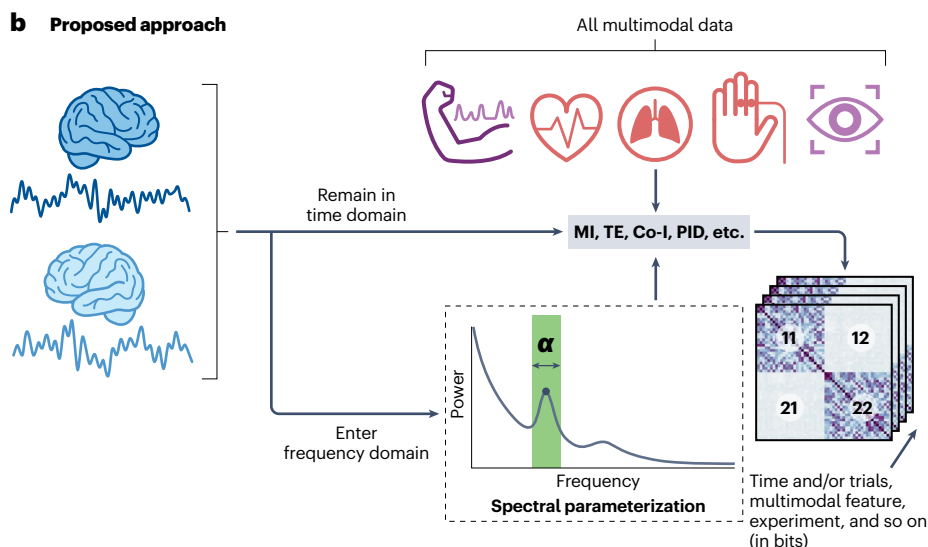
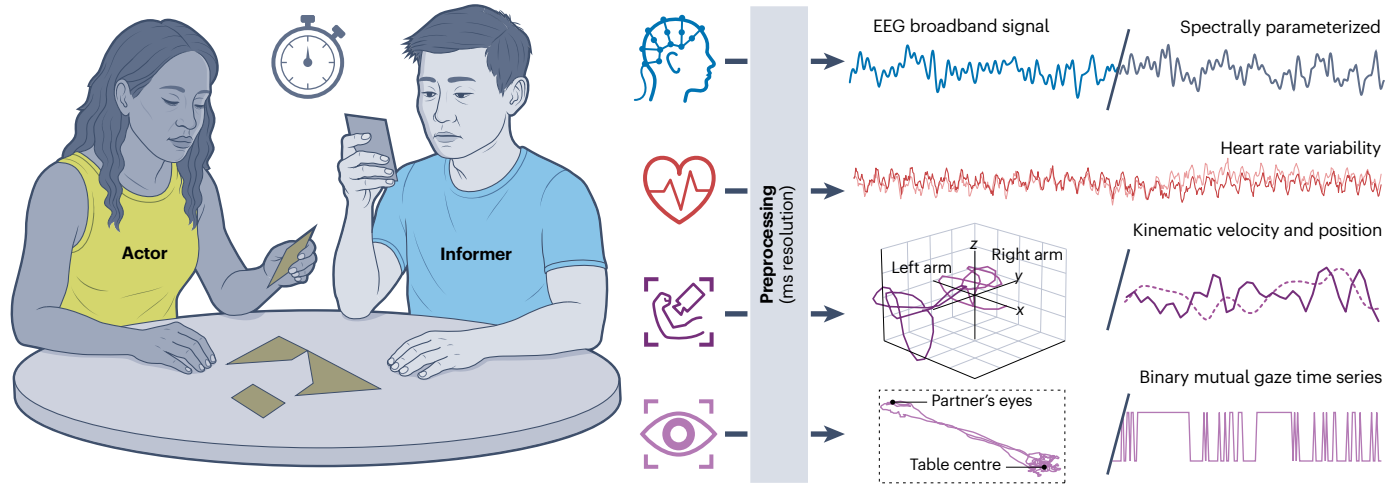


Fig. 4 | Approaches to studying multimodal interpersonal coordination. **a**, (1) Current approaches use band-pass filters to transform neural time-series signals into canonical frequency bands of interest (such as alpha (α ; 8–13 Hz), often determined a priori) and then apply various standard metrics including phase locking value (PLV), phase lag index (PLI), Granger causality (GC) and coherence, among others. Hyper-matrices of pairwise interactions (such as between electrodes) can be generated (right), wherein 11 refers to the intra-dynamics of person 1, 12 refers to the directed inter-dynamics of person 1 to 2, and so forth. However, this can be computed only across time or trials. (2) Following ref. 55, interpersonal interactions can be modelled as a cross-brain general linear model, that is, as a function of (or linear combination of) multiple, independent within-person and across-person predictors (including neural, behavioural, physiological) and task-related (such as collaborative versus competitive conditions) data.

b, In the proposed information-theoretic approach, researchers can bypass band-pass filtering and maintain broadband signals in the time domain. Equally, spectral parameterization of one's neural data should inform the use of frequency-resolved and even time–frequency-resolved information-theoretic measures. These information-theoretic measures include mutual information (MI), transfer entropy (TE), Co-I (or interaction information) and partial information decomposition (PID), among others. Moreover, any and all multimodal data can be included in the various information-theoretic analyses (such as conditioning on a stimulus or analysing between modalities simultaneously), generating hyper-matrices across various domains given in bits.

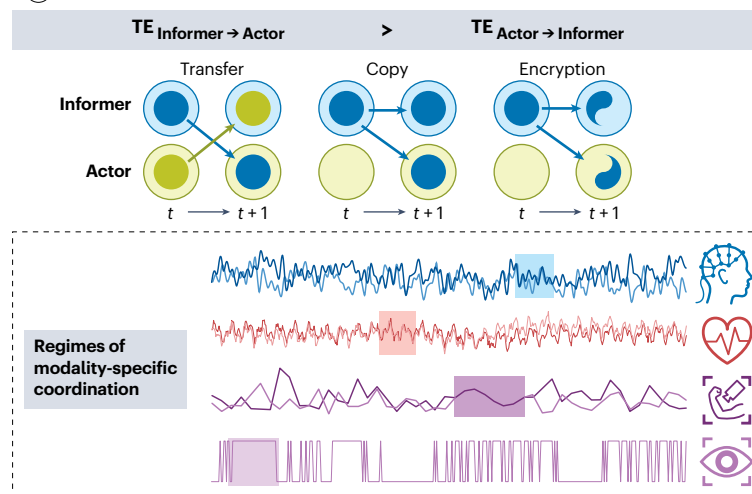
Perspective

a Experimental set-up and multimodal data

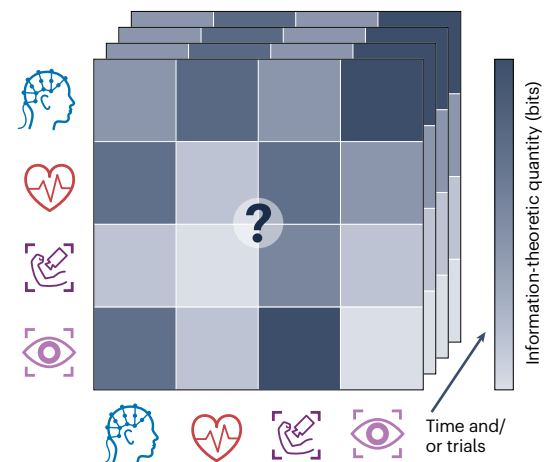


b Information-theoretic analyses

1 Testable predictions

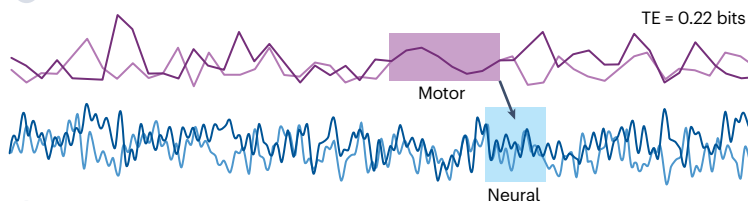


2 Interpersonal multimodal coordination matrix

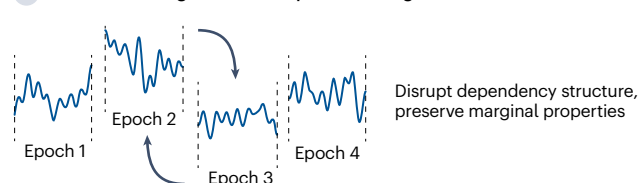


3 Statistical inference

1 Calculate test statistic



2 Generate surrogates with temporal shuffling



3 Empirical null distribution via Monte Carlo shuffling

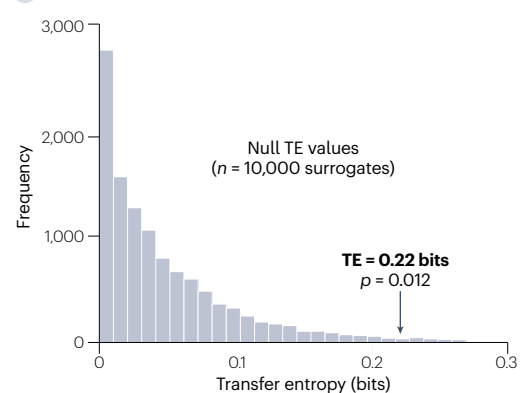


Fig. 5 | Example of a multimodal hyperscanning experiment. **a**, Dyads jointly complete a cooperative spatial puzzle (such as tangram construction) in which one participant (the informer) holds the target image and another participant (the actor) manipulates pieces to recreate the shape. Duration of task completion is timed. Simultaneously, four data modalities are recorded at millisecond resolution: neural signals (electroencephalography (EEG)) kept either at broadband or spectrally parameterized, autonomic physiology (heart rate variability), kinematics (upper-limb motion via markerless tracking), and gaze behaviour (binary mutual gaze or gaze coordinates). These signals are preprocessed and synchronized. **b**, (1) Nonlinear information-theoretic metrics enable hypothesis-driven analyses of dynamics of interpersonal coordination. For instance, directional measures such as transfer energy (TE) might reveal that more information is transferred from the informer to the actor than vice versa, corresponding to role-driven dynamics. High-order qualitative informational constructs such as transfer, copy or encryption can also be explored through interaction information and co-information. Furthermore, time-resolved decompositions can uncover modality-specific coordination regimes (shaded boxes) such as gaze-led or motion-led episodes. (2) All pairwise combinations of

signals between partners can be assessed using information-theoretic measures such as mutual information or TE. This generates a multimodal coordination matrix with entries in bits, allowing direct comparison across data types (akin to synchronization matrices⁵⁹). Trials can be analysed either as single events, assuming stationarity, or measures can be deployed per time point across trials to accommodate non-stationary dynamics. The values in this schematic matrix are left ambiguous as they remain open empirical questions. (3) To assess the significance of observed interpersonal coordination, Monte Carlo permutation testing is used. First, a test statistic is computed (for example, TE = 0.22 bits), which may include the median information-theoretic quantity across trials as in this example (top, left). Next, surrogate data sets are generated by temporally shuffling the time series of one participant, destroying temporal dependencies while preserving the marginal distributions (bottom, left). An empirical null distribution of TE values is then constructed over many iterations (for example, $n = 10,000$), against which the original value is compared to yield an empirical p value (right). This procedure ensures observed interdependence exceeds what would be expected from shared stimulus timing or coincidental dynamics.

For the linguistic dimension, it is important to note that information-theoretic approaches, as formulated here, are agnostic to the semantic content of communication. The framework of Shannon quantifies the statistical dependencies between variables rather than explaining how interactants establish shared meanings or interpret referents¹⁶⁷. Given the context-dependency and idiosyncratic nature of linguistic and emotional communication^{168–170}, information theory is most applicable to studying interpersonal coordination in structured, multimodal exchanges (such as joint action, turn-taking and shared affect) rather than analysing how meaning is constructed in conversation. Future work could explore how information-theoretic tools might be integrated with pragmatic or inferential models of communication (or even representational analyses) to directly encode the information content⁷⁴. Emerging works using large language model embeddings, to numerically approximate the shared meaning space that interlocutors draw upon when expressing and interpreting thoughts¹⁷¹ and when measuring conversational alignment on lexical, conceptual and syntactic levels^{172,173}, are certainly important steps towards achieving this.

Although information theory has existed for decades, the adoption of its methods in social psychology and neuroscience has been constrained owing to the limited availability of closed-form solutions for information-theoretic quantities (that is, formulas that can be solved exactly rather than estimated through simulations or numerical methods), thus requiring data-hungry and computationally intensive estimators: a necessary trade-off to avoid assumptions such as linearity and Gaussianity. Given that data and computing power have become plentiful in recent years, and new algorithmic developments have further improved efficiency, the time is now ripe to capitalize on its advantages¹⁷⁴. It is important to note that although information-theoretic analyses can require more samples than simple linear approaches, high-temporal-resolution modalities such as EEG and optically pumped magnetometer-magnetoencephalography offer a promising solution by providing rich, high-volume data sets suitable for estimating these quantities with fewer participants^{128,175}. Nonetheless, appropriate choice of algorithms and parameters is necessary to balance bias, computation time and data-efficiency, complicating the deployment of information-theoretic measures and inevitably influencing final estimations^{128,176}. For example, multiple debiasing methods are available and continually emerging^{141,177–180}. Indeed, information theory itself is an active area of research. Although this expanding

plethora of options allows researchers to choose the most appropriate implementation for their specific needs, we acknowledge that it can be challenging for newcomers and may continue to heterogenize analytical approaches. To this end, readers are directed to multiple well-recognized software packages and tutorials^{13,128,141,178,181–187}.

In any case, despite its solid foundations in most areas of neuroscience, information theory has been conspicuously absent from social neuroscience, with just a handful of recent applications that we are aware of^{68,129–138}. Indeed, given recent advances in both data collection and computing resources, the study of interpersonal coordination can now certainly meet the once-demanding requirements of the proposed methods; it is time to depart from inexpensive but limited correlations and embrace information-theoretic methods (Fig. 4).

Together, the proposed framework provides a principled foundation for quantifying information dynamics during social interaction. Building on this, we now outline specific empirical predictions that follow from applying these measures to multimodal hyperscanning data.

Predictions

Beyond theoretical insights, the proposal framework enables empirical tests with specific predictions. First, nonlinear dependencies may prove to better capture interpersonal coordination processes (such as movement adaptation or social gaze following), whereas nonlinear models could more accurately classify social interaction states such as turn-taking or joint problem-solving, whether (non-) successful, cooperative or competitive. Indeed, this may similarly extend to discoordination among interactants with social dysfunctions and psychopathologies^{2,188–190}. Second, leader–follower dynamics may involve asymmetric information flow (identifying information senders and receivers), detectable with TE rather than coherence or autoregressive approaches. Third, time-resolved measures can reveal metastable dynamics of interpersonal coordination, showing transient shifts between alignment and independence that are often missed by conventional methods. Last, higher-order information sharing (particularly Rdn and Syn encoding) could test whether multimodal variables (such as gaze and hand motion, or neural activity across regions) encode shared information about a behavioural outcome (such as task duration) or experimental condition (such as strangers versus friends, teacher–student versus student–student, or therapist–client). Here, Rdn would indicate overlapping encoding (perhaps reflecting an

attempt to ensure that information is correctly conveyed by not relying on a single, potentially flawed channel), whereas Syn could reveal how combined fluctuations in physiology, gestures and gaze encode shared moments of decision-making better than any modality alone could.

One parent–child EEG hyperscanning study has already demonstrated striking differences in the sensitivity of standard phase-based and information-theoretic measures¹⁹¹. In interactive social learning and affective decision-making tasks with 80 dyads, only nonlinear connectivity (as measured by MI) captured maturational (age) changes and showed task-relevant modulation by infants. These findings suggest that over development, a shift in dominance from linear to nonlinear forms of information processing may be essential for supporting emerging precursors to higher cognitive abilities such as executive function (for example, attention shifting) and social decision-making. Although the study examined within-brain connectivity, implications extend for hyperscanning: if nonlinear metrics better track cognitive engagement and developmental progression within individuals, they may also yield more robust insights when applied across interacting brains, wherein complexity and mutual influence may amplify these dynamics. Furthermore, recent fMRI investigations using information decomposition have demonstrated that conventional connectivity indices based on synchronous co-fluctuation (correlation, covariance) primarily reflect Rdn information and overlook Syn information^{160–163}. How these classes of measure compare in hyperscanning studies remains an open question for analysis of existing data sets, new empirical designs and even hyper-brain simulations (computational models that simulate multiple interacting brains; see ref. 86 for an example).

An illustrative experimental set-up (Fig. 5a) could involve dyads jointly solving a spatial puzzle, such as reconstructing abstract patterns with tangram pieces (see ref. 49 for a comparison). One participant (the ‘informer’) views the target image, whereas the other participant (the ‘actor’) manipulates the pieces, requiring interpersonal coordination through verbal and nonverbal means. Multimodal data are simultaneously recorded, using EEG for neural activity, eye-tracking for gaze behaviour, motion capture for upper-body kinematics, and wrist sensors for heart rate and electrodermal activity. Because the signals in each modality operate on the millisecond timescale, they can be synchronized to a common temporal resolution without substantial interpolation, unlike fMRI and fNIRS, wherein slower sampling may necessitate downsampling or upsampling. Indeed, the modality with the lowest temporal resolution will remain the bottleneck (such as kinematics; Fig. 5a). Nonetheless, such disparities in temporal resolution may map onto a distinction between alignment of states (slower dynamics; for example, turn-taking or state changes) and alignment of content (fine-grained; for example, exchange of information content). To control for shared acoustics in neural dynamics, speech envelopes can be regressed out of the EEG signal¹⁹², or silence could be enforced. Finally, effective interpersonal coordination could be operationalized behaviourally with, for example, time to puzzle completion, number of object misplacements or movement efficiency (such as trajectory smoothness or distance travelled by the hands of the actor).

This design lends itself to various mechanistic questions: do sustained interpersonal coordination episodes (single modality, cross-modal, multimodal) reliably precede correct piece placement and faster completion (proxies of effective coordination)? Do they coincide with heightened inter-brain coordination in particular neural sources (such as in frontoparietal or so-called social brain regions) in either the actor or the informer? Additionally, what is the directionality

of interpersonal coordination; that is, does information flow predominantly from informer to actor, and through which modality (gaze, gestures)? Information-theoretic approaches can address these questions (as outlined in Fig. 5b).

Although information theory may afford a universal statistical language and mathematical framework for analyses, interpreting the seemingly abstract unit of bits still requires that it be grounded within the experimental context. For example, Becchio and colleagues¹⁹³ show how kinematic encoding-readout models translate MI between movement features and intentions into a common representational space for both neural and behavioural data, allowing statements about when and how intentions become available for social prediction. Likewise, Combrisson and colleagues¹⁹⁴ use PID of MEG signals to reveal that information gain during goal-directed learning is carried by Syn higher-order interactions centred in prefrontal cortex hubs, identifying network mechanisms for belief updating and broadcast. In this way, numerical outputs become directly relevant to theories of coordination, prediction and mutual adaptation; in other words, the mathematics remain anchored in mechanistic accounts of social interaction.

However, testing the causal role of interpersonal coordination requires going beyond observation to carefully controlled intervention. Although hyperscanning has revealed correlational links between inter-brain coordination and social behaviour, causal evidence remains contested and unresolved. To this end, multi-brain stimulation offers promise in both humans (for example, via hyper-transcranial alternating current stimulation^{195–197}) and animal models (for example, via optogenetics^{198,199}) to assess its effect on interpersonal coordination and downstream social behaviour. Stimulation protocols replicating empirically observed interpersonal coordination (rather than the simple sine-wave simulation from transcranial alternating current protocols) may better approximate natural neural dynamics, including their consequences on behavioural and physiological scales, thereby enhancing ecological validity⁷⁵. Complementary methods, including virtual partner paradigms and computational modelling, can refine explanatory models and even inform stimulation design⁷⁶. Although direct intervention largely remains the gold standard for causal inference²⁰⁰ (but see refs. 201,202 for alternative frameworks for achieving causal inference), analytical tools for statistical causal discovery, such as TE and multivariate information decomposition, can clarify directionality and interdependence (although not conclusively establish causality)^{142,156,203}. Beyond their methodological value, such approaches might help to adjudicate ongoing debates about the role of multi-brain stimulation as a causal probe in social neuroscience. Critically, interventions should extend to multimodal interpersonal coordination, too – for example, experimental manipulation of visual fields (introducing latency while viewing one’s partner digitally), motion or physiological synchrony. Similarly, virtual partners or avatars with parametric control over their features offer another powerful tool for causal probing.

Together, these predictions outline a concrete research agenda for testing how information-theoretic measures can reveal the dynamics, directionality and causal mechanisms of interpersonal coordination across modalities and contexts. By integrating nonlinear modelling, higher-order information decomposition, and experimental intervention, this framework moves beyond correlation to mechanistic explanation. In the next section, we turn to the broader implications of this shift and consider how embracing nonlinear, data-driven approaches can pave a more unified and rigorous way forward for the study of multimodal interpersonal coordination.

Glossary

Coherence

A measure of the consistency of phase and amplitude relationships between two signals over time and frequency.

Cross-frequency interactions

Dependencies between neural rhythms at different frequencies (for example, phase–amplitude or phase–phase relations) hypothesized to coordinate information processing across temporal scales.

Dynamical systems

Systems whose states evolve over time with interdependent components, often exhibiting nonlinear, context-sensitive and multiscale behaviour; used here for both neural and social processes.

Emergent

A property or behaviour arising from interactions among components that cannot be reduced to component-level properties alone.

Entropy

In information theory, entropy (uncertainty) describes how unpredictable a system or variable is; it reflects the average surprise or information gained when observing an outcome: systems with many equally probable outcomes have high entropy, whereas those with predictable outcomes have low entropy.

Granger causality

A directional time-series framework in which X ‘Granger-causes’ Y if past values of X improve prediction of future values of Y beyond Y ’s own history.

Hyperscanning

Simultaneous recording of neural activity from two or more interacting individuals, using comparable neuroimaging or electrophysiological modalities, to investigate brain-to-brain coordination during actual social interaction.

Information

In the information-theoretic sense, information is quantified reduction in uncertainty about one variable gained by observing another, measured in bits; the common currency of the information-theoretic framework adopted here.

Integrated information decomposition

An extension of partial information decomposition that applies to dynamic systems with multiple sources and targets, tracking how unique, redundant and synergistic information evolve over time to characterize processes such as information storage, transfer and integration.

Metastability

A dynamic regime in which a system transiently occupies multiple quasi-stable coordination states, switching between periods of alignment and misalignment over time.

Multi-brain stimulation

Experimental approach that simultaneously delivers controlled neural stimulation to two or more interacting individuals, typically using techniques such as hyper-transcranial alternating current stimulation in humans or optogenetics in animal models.

Multimodal interpersonal coordination

Dynamic alignment between interactants across multiple levels of organization — neural, behavioural, physiological, affective and linguistic — during real-time social engagement.

Mutual information

A symmetric information-theoretic measure quantifying the statistical dependence between two variables; it represents the reduction in uncertainty about one variable gained by knowing the other, capturing both linear and nonlinear relationships.

Mutual prediction hypothesis

The view that inter-brain similarity arises because individuals independently predict and adapt to each other’s behaviour, without requiring direct neural communication across brains.

Nonlinear

A relationship in which outputs are not proportional to inputs and effects depend on context or interactions, resisting additive, fixed-form linear modelling.

Partial information decomposition

A framework for breaking down the total information that multiple source variables provide about a target variable into distinct components — unique, redundant and synergistic information — thereby revealing how variables jointly or individually contribute to explaining the target.

Phase

The position of a signal within its oscillatory cycle; in neuroscience, phase relationships between signals can indicate the timing alignment or synchrony of neural oscillations.

Power spectral density

A representation of how signal power is distributed across frequencies, used to characterize oscillatory peaks.

Redundant information

Information about a target that is available from multiple sources simultaneously, such that it would remain even if one of the sources were removed; reflects overlapping encoding among predictors.

Synchrony

A time-locked subset of coordination referring to simultaneous or phase-aligned activity between signals, contrasted with broader forms of coordination that need not be strictly simultaneous.

Synergistic information

Information available only from the joint consideration of multiple sources and not from any one source alone, revealing higher-order dependencies.

Transfer entropy

A directional, model-free measure of predictive influence from X to Y , defined as the conditional mutual information between X and the next state of Y given Y ’s past; it quantifies information in X about Y that is not contained in Y ’s own history (equivalent to Granger causality under linear-Gaussian assumptions).

Unique information

The portion of information about a target that is provided exclusively by one source and not shared with others, isolating the distinctive informational contribution of that source within a system.

Paving a nonlinear way forward

With the rise of multimodal experiments composed of multiple levels of analysis in the study of interpersonal coordination⁵, developing a unified and data-driven framework for quantifying and qualifying the coordination of the informational dynamics of social systems has become critical. We must face the possibility that these multimodal interpersonal systems cannot be written as a linear sum of independent

components; a unified statistical language for capturing potential nonlinearities and interpreting experimental findings is essential. Indeed, the use of data-driven methods with less-restrictive assumptions and ad hoc variable selection is now possible owing to the greater availability of both data and computing power. These approaches will not only assist in capturing any nonlinearity and higher-order interactions in both temporal and spectral domains, but also enable ways to more faithfully

describe the statistical properties of interpersonal coordination data and the complexity of data interactions. Nonetheless, we emphasize the importance of appropriate methodological decisions when deploying information-theoretic quantities and when providing accessible tutorials and pipelines, given that multiple estimation methods and parameter choices exist, and their misuse or misinterpretation can easily lead to divergent or unreliable results.

In this Perspective, building on the promises of the study of interpersonal coordination, we have sought to address current challenges in this rapidly evolving field. We have proposed information theory as a mathematical framework capable of handling cross-modal features while capturing complex, potentially nonlinear phenomena on both individual and collective scales. In all, we believe that this approach can make a critical contribution to the analyses of the field that will ultimately enable researchers to better substantiate emerging theories of multimodal interpersonal coordination.

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